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Paleomagnetic evidence for Neogene clockwise tectonic rotations in the central Mojave Desert, California

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ABSTRACT

Paleomagnetic directions from early Miocene volcanic flows indicate that a large part of the central Mojave Desert has been tectonically rotated. Stratigraphic relations and isotopic ages suggest that two separate clockwise rotations have occurred. An early Miocene rotation of 50° ±15.6° is coincident in time and space with opening of the Daggett extensional terrane. A post–early Miocene rotation of 30° to 50° may have affected the Alvord Mountains and the immediate vicinity.

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INTRODUCTION

The Mojave Desert block (Dokka, 1983) of southeastern California is bounded on the west by the San Andreas fault and on the north by the Garlock fault. This region is crossed by many faults that are either northwest-striking, subparallel to the San Andreas fault, or east-striking, subparallel to the Garlock fault. The

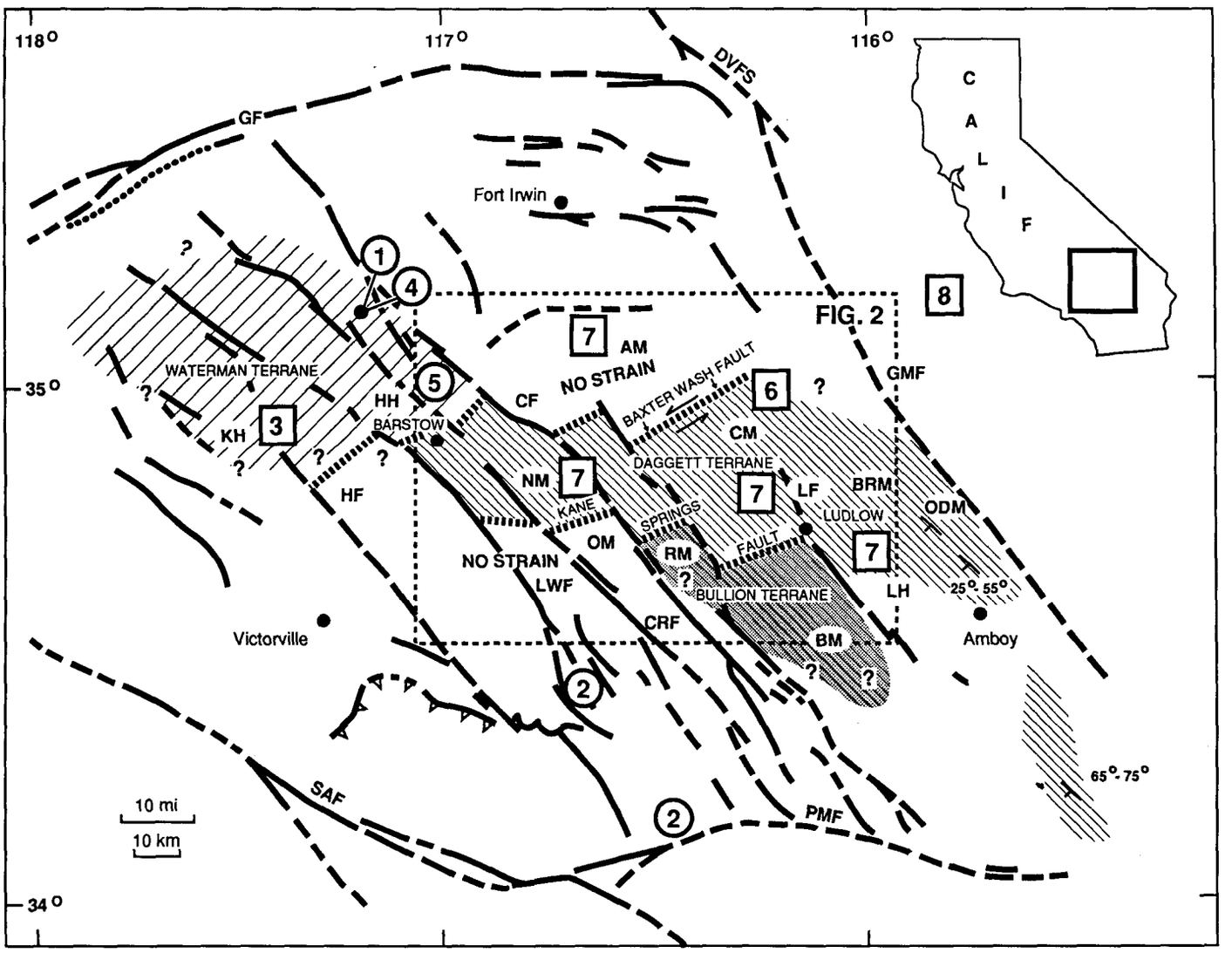


Figure 1. Generalized map of Mojave Desert showing study area, fault trends, early Miocene extensional terranes of Dokka and Woodburne (1986), and previous paleomagnetic studies. Squares indicate studies yielding clockwise paleomagnetic directions, circles indicate counterclockwise (modified from Dokka and Woodburne, 1986). 1—Burke et al. (1982); 2—Morton and Hillhouse (1983, unpublished); 3—Golombek and Brown (1988); 4—Valentine et al. (1987, 1988); 5—MacFadden et al. (1987, 1988); 6—MacFadden et al. (1989); 7—Ross et al. (1987, 1988); 8—Harlan and Geissman (1987). AM = Alvord Mountains, CM = Cady Mountains, BRM = Bristol Mountains, BM = Bullion Mountains, OM = Ord Mountains, NM = Newberry Mountains, RM = Rodman Mountains, ODM = Old Dad Mountains, KH = Kramer Hills, HH = Haloran Hills. GF = Garlock fault, CF = Calico fault, GMF = Granite Mountains fault, LF = Ludlow fault, DVFZ = Death Valley fault zone, CRF = Camp Rock fault, LWF = Lenwood fault, HF = Helendale fault, SAF = San Andreas fault, PMF = Pinto Mountain fault.

latter faults are particularly evident in the north-central part of the province. The geometry of these faults has inspired tectonic models by Garfunkel (1974), Luyendyk et al. (1980), and Carter et al. (1987) that predict crustal rotations about a vertical axis in the Mojave Desert block.

Paleomagnetic studies in Neogene-age rocks have provided evidence for tectonic rotations throughout this province. Valentine et al. (1987, 1988) have reported counterclockwise deflected paleomagnetic declinations in the central Mojave (Fig. 1). Studies by Burke et al. (1982), MacFadden et al. (1987, 1988; 1988, personal commun.), and Morton and Hillhouse (1983, unpublished) have observed both clockwise and counterclockwise deflected declinations. Luyendyk et al. (1985), Golombek and Brown (1986, 1988), Ross et al. (1987, 1988), and Harlan and Geissman (1987) have reported clockwise deflected declinations in the Mojave Desert block (Fig. 1). The purpose of this study is to document and quantify the amount, direction, and timing of tectonic rotations in the central Mojave Desert block.

GEOLOGIC SETTING

During late Oligocene and early Miocene time, andesitic and basaltic lavas erupted through a basement of Mesozoic and older igneous and metamorphic rocks contemporaneous with an episode of crustal extension (Armstrong and Higgins, 1973; Dokka and Woodburne, 1986). Isotopic dates from correlative volcanic units in this area are generally between 20 and 26 Ma (Armstrong and Higgins, 1973; Dokka, 1986; Dokka and Woodburne, 1986; Nason et al., 1979).

Extension in the central Mojave Desert was associated with low-angle normal faulting, producing "domino style" fault-block relations (Dokka, 1986; nomenclature after Wernicke and Burchfiel, 1982). The main direction of extension was northeast-southwest (in present coordinates), causing most bedding attitudes to dip to the southwest and faults to dip steeply northeast (Dokka, 1986). After the main phase of extension, strata correlated with the regionally extensive Peach Springs Tuff (Glazner et al., 1986) were deposited over tilted flows and beds at about 18 to 19 Ma (Glazner, 1986). Continental sediments were then deposited over most of the central Mojave Desert through middle Miocene time (Miller, 1980). After a period of quiescence, northwest-trending dextral shear became the dominant tectonic force in the Mojave Desert. Onset of northwest-striking right-lateral faulting may have begun as early as 13 Ma (Dokka, 1988).

PALEOMAGNETIC RESULTS

A total of 108 paleomagnetic sites composed of 7 or more oriented samples have been collected in 92 early Miocene volcanic flow units from the central Mojave Desert (Fig. 2). A gasoline-powered drill was used to take 2.5-cm-diameter cores, which were oriented in the field to ±2° by using both magnetic-compass and sun-compass techniques. A Molspin spinning fluxgate magnetometer was used to determine the remanent magnetic direction held by the samples during alternating field (AF, up to 90 mT) and stepwise thermal (SWT, up to 650 °C) demagnetization. These paleomagnetic direc-

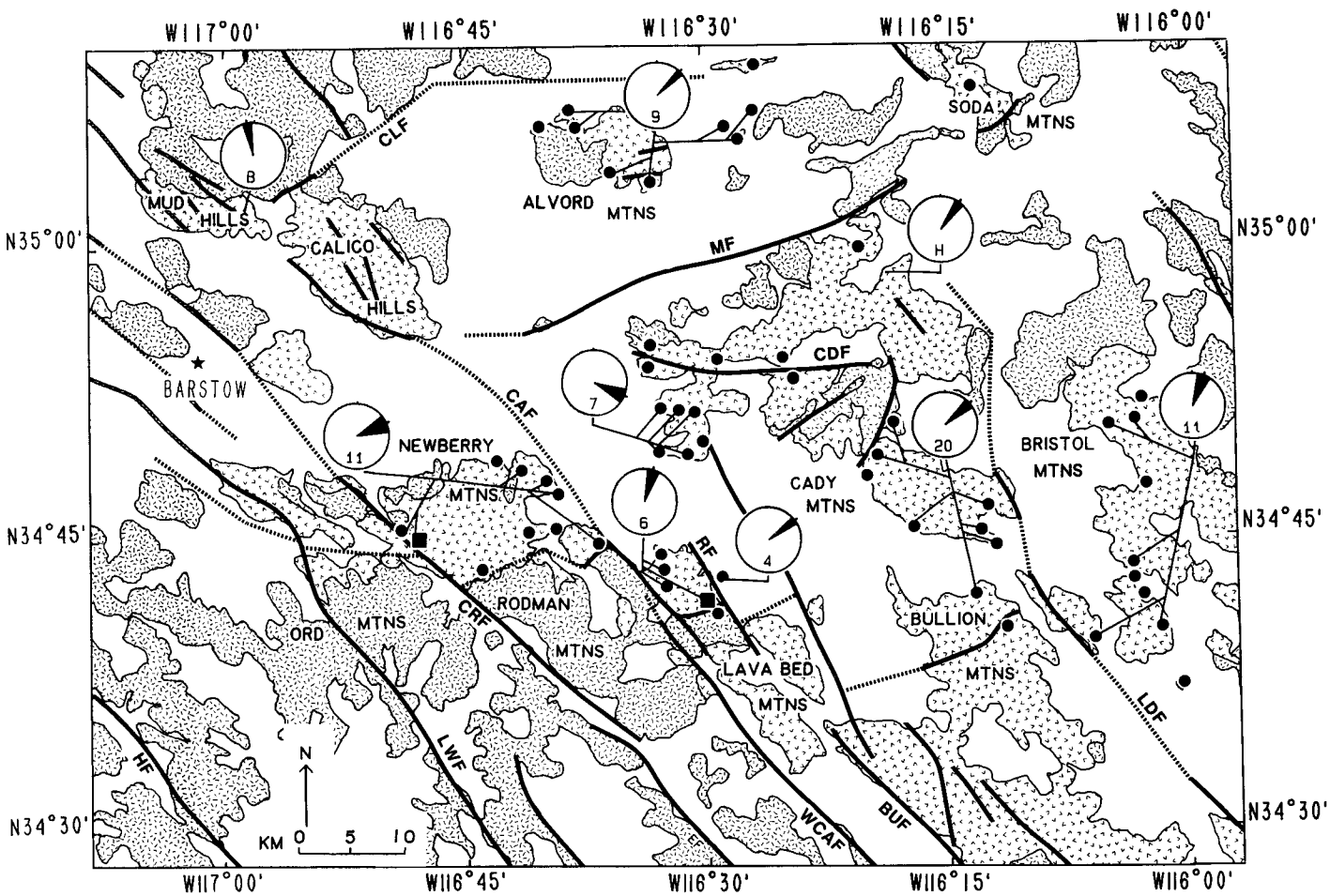


Figure 2. Generalized geologic map of central Mojave Desert (after Dokka, 1986; Jennings et al., 1963; and Rogers, 1969). V pattern = Tertiary volcanic rocks; random-dash pattern = pre-Tertiary rocks, mostly granites of Mesozoic age. Solid circles = paleomagnetic sampling locations from this study; squares = locations from Morton and Hillhouse (1983, unpublished). Mean declinations for individual structural blocks are indicated with their associated 95% confidence. Number of sites averaged and their locations are shown. CLF = Coyote Lake fault, MF = Manix fault, CDF = Cady fault, CAF = Calico fault, HF = Helendale fault, LWF = Lenwood fault, RF = Rodman fault, WCAF = West Calico fault, BUF = Bullion fault, LDF = Ludlow fault.

tions have been corrected for structural tilting by using local bedding attitudes. Wherever possible, the structural attitude of nearby or interbedded sedimentary rock was used for the tectonic correction to help decrease any error caused by untilting a flow that was deposited with an initial dip.

Stable characteristic directions were picked by analyzing stereographic projections, vector end point plots (Zijderveld, 1967), and difference vectors from the progressive demagnetization of each sample. The stable directions from the samples in each site have been averaged to form a site mean. A site mean direction is determined to be acceptable if the α_{95} is less than 15°. Of the 108 sites, 74 display stable and acceptable directions. In order to account for the effects of secular variation of Earth's magnetic field, these site directions have been averaged together in groups from structurally bounded crustal blocks. These structural blocks correspond to individual mountain ranges, except in the Cady Mountains where the paleomagnetic data support the subdivision of this area using the Pisgah and Cady faults as boundaries.

The inclination of these average paleomagnetic directions moves closer to the expected inclination and the dispersion decreases (α_{95} decreases, K increases) with application of structural corrections (Table 1; Fig. 3). These two observations suggest that the magnetic directions held by these rocks were captured prior to early Miocene tilting of the volcanic units. The mean directions from these blocks are all deflected clockwise from both the present axial geocentric dipole direction (true north), and the early Miocene reference direction (I = 53.3, D = 351.8, calculated from the early Miocene reference pole of Diehl et al., 1983; lat = 83.2, long = 148, α_{95} = 4.1). The average direction for these blocks (excluding the anomalous west Cady Mountains and the possibly younger Alvord Mountains results; Table 1) is I = 56.8, D = 41.7 (α_{95} = 10.5, K = 41.8). We interpret the paleomagnetic data to indicate that these rocks have been rotated approximately 50° clockwise about a vertical axis sometime after early Miocene time.

DISCUSSION

A flat-lying outcrop correlated to be the Peach Springs Tuff (Glazner et al., 1986) overlies tilted and rotated early Miocene strata in the southern Newberry Mountains. The paleomagnetic direction held by this outcrop is not significantly different from the direction for the Peach Springs Tuff located on the stable Colorado Plateau (Hillhouse and Wells, 1986; Wells and Hillhouse, 1989). This comparison suggests that clockwise rotation in the Newberry Mountains was finished by 18 to 19 Ma when the Peach Springs Tuff was deposited (Glazner et al., 1986; Nielson et al., 1988). A paleomagnetic study in the Cady Mountains north of the Cady fault suggests that 23 to 16 Ma rocks of the Hector Formation have rotated clockwise about 26.7° since 16 Ma (MacFadden et al., 1989). This suggests that the northern Cady Mountains rotated clockwise after rotation stopped in the Newberry Mountains to the south. Therefore, there may have been two distinct periods of clockwise rotation in the central Mojave Desert; one period in early Miocene time and one in post–early Miocene time.

The models of Garfunkel (1974), Luyendyk et al. (1980), and Carter et al. (1987) assume that movement on northwest-striking right-slip faults would accompany counterclockwise or clockwise rotation of crustal blocks. The northwest-striking faults in the Mojave Desert developed after 13 Ma, and they have probably been most active since 5 Ma (Dokka, 1983, 1988). Therefore, these models cannot adequately explain the early Miocene clockwise rotation in the central Mojave Desert. However, all our paleomagnetic studies except for the Alvord Mountains lie within the boundaries of the Daggett extensional terrane active from about 22 to 18 Ma (Dokka, 1986, and in prep.). Because the clockwise rotation and the tectonic extension of the same area appear synchronous, we suggest that a large area of the central Mojave Desert extended and rotated clockwise in a fairly coherent manner between about 22 and 18 Ma.

The northern Cady Mountains are located south of the Alvord Mountains and separate the Alvord Mountains from the Daggett extensional terrane. We interpret the Alvord Mountains to have rotated in post–early Miocene time along with the northern Cady Mountains. The southern boundary of this post–early Miocene rotation is presumed to be the Cady fault, following Luyendyk et al. (1980). The Alvord Mountains and the north Cady Mountains are bounded by east-striking to northeast-striking faults (Fig. 2) and are the crustal blocks that the Carter et al. (1987) model predicts should have been rotated clockwise in post–mid Miocene time. Revised isotopic dates in the eastern Transverse Ranges (Paul Stone, 1988, personal commun.) constrain the rotation found here by Carter et al. (1987) to be post–6 Ma. That rotation of the eastern Transverse Ranges occurs simultaneously with the rotation of the Alvord and north Cady mountains, according to the Carter et al. (1987) model, suggests that the Alvord and north Cady mountains have rotated post–6 Ma.

The west Cady Mountains exhibit magnetic declinations that are much more strongly clockwise deflected than those in the rest of our study area (Fig. 2; Table 1). The west Cady Mountains may have undergone a much larger rotation than the rest of the Daggett terrane in early Miocene time. Alternatively, the western Cady Mountains may have undergone two different clockwise rotations: one in early Miocene time along with the rest of the Daggett terrane,

TABLE 1. DISCORDANCE STATISTICS, CENTRAL MOJAVE DESERT, CALIFORNIA

Block	Inc. (°)	Dec. (°)	N	α_{95}	K	R	ΔR	F	ΔF
Newberry	49.5	64.5	11	20.1	9.2	72.7	20.0	3.8	12.7
Rodman	49.6	17.3	6	13.9	39.0	25.5	13.6	3.6	8.7
Lava Bed	65.3	53.9	4	9.2	175.0	62.5	13.6	-12.0	5.6
Bristol	59.1	21.5	11	15.7	15.0	29.6	19.8	-6.0	9.9
South Cady	50.5	48.5	20	11.6	14.0	56.6	11.6	2.7	7.4
West Cady*	49.2	115.7	7	16.8	22.0	124.0	16.2	4.1	10.5
Alvord†	59.1	45.0	9	8.0	69.0	53.2	9.9	-6.0	5.0
Average 1§	56.8	41.7	6	10.4	41.8	49.8	15.6	-4.0	8.4
Average 2§	56.3	41.1	5	13.1	33.9	49.2	19.6	-3.0	10.6
Average 3§	54.1	43.5	52	5.9	12.0	51.6	8.2	-1.0	4.8

Note: N =number of samples averaged; α_{95}, K are statistical parameters of Fisher (1953); R, ΔR, F, ΔF are rotation and flattening; 95% confidence levels calculated by method of Beck (1980) modified by Demarest (1983). Reference directions: inclination = 53.3, declination = 351.8 (computed from the Miocene reference pole of Diehl et al., 1983; lat = 83.2, long = 148.0, α_{95} = 4.1).

* Anomalous direction: not used in calculations of averages.

† Not used in calculations of averages 2 and 3 due to timing of rotation.

§ Average 1: from structural blocks without West Cady Block; Average 2: structural block directions from Daggett terrane; Average 3: site directions from Daggett terrane.

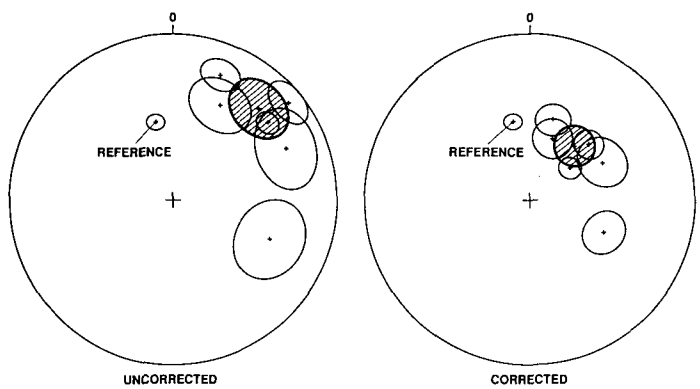


Figure 3. Mean directions from each structural block plotted on equal-area stereonet with their associated 95% confidence levels. All mean directions are deflected clockwise from present geomagnetic reference direction (north) and Miocene reference direction of Diehl et al. (1983) shown. Average direction of areas interpreted to have been rotated in early Miocene time is indicated by shading (does not include West Cady or Alvord mountains; if Alvord Mountains are included, mean direction is I = 30.3, D = 45.4, α_{95} = 18.7 (no tilt correction), I′ = 56.8, D′ = 41.7, α_{95} = 10.5 (corrected for tilt).

and a subsequent rotation in post–early Miocene time along with the north Cady and Alvord mountains. Due to the large amount of rotation inferred to have affected the west Cady Mountains (124°), we favor the two-rotation hypotheses.

CONCLUSIONS

The regional consistency of the paleomagnetic data presented here suggests that a large part of the central Mojave Desert has undergone clockwise tectonic rotation since Oligocene time. Timing constraints suggest that there have been two different episodes of clockwise rotation. A post–early Miocene clockwise tectonic rotation appears to have affected the north Cady, Alvord, and possibly the western Cady mountains.

The location and timing of the early Miocene clockwise rotation suggested by our data coincide with early Miocene extension in the Daggett terrane which encompasses the Newberry, Rodman, Lava Bed, Bristol, south Cady, and west Cady mountains of this study. Golombek and Brown (1986, 1988) presented clockwise rotated paleomagnetic directions from early Miocene strata located in the Waterman extensional terrane (Dokka and Woodburne, 1986; Fig. 1). This correlation between the rotation and extension leads to these testable hypotheses: (1) An area of tectonic rotation exists that is coincident with the boundaries of the Daggett terrane which shows the same sense and same amount of rotation. (2) Other domains of early Miocene extension (the Waterman and Bullion terranes of Dokka and Woodburne, 1986) may also show associated coherent clockwise tectonic rotations.

REFERENCES CITED

Armstrong, R.L., and Higgins, R.E., 1973, K-Ar dating of the beginning of Tertiary volcanism in the Mojave Desert, California: Geological Society of America Bulletin, v. 84, p. 1095–1100.

Beck, M.E., 1980, Paleomagnetic records of plate margin tectonic processes along the western edge of North America: Journal of Geophysical Research, v. 85, p. 7115–7131.

Beck, M.E., Burmester, R.F., Craig, D.E., Gromme, C.S., and Wells, R.E., 1986, Paleomagnetism of middle Tertiary volcanic rocks from the Western Cascade Series, northern California: Journal of Geophysical Research, v. 91, p. 8219–8230.

Burke, D.B., Hillhouse, J.W., McKee, E.H., Miller, S.T., and Morton, J.L., 1982, Cenozoic rocks in the Barstow Basin area of southern California—Stratigraphic relations, radiometric ages, and paleomagnetism: U.S. Geological Survey Bulletin 1529-E, 16 p.

Carter, J.N., Luyendyk, B.P., and Terres, R.R., 1987, Neogene clockwise rotation of the eastern Transverse Ranges, California, suggested by paleomagnetic data: Geological Society of America Bulletin, v. 98, p. 199–206.

Demarest, H.H., Jr., 1983, Error analysis for the determination of tectonic rotation from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.

Diehl, J.F., Beck, M.E., Jr., Beske-Diehl, S., Jacobson, D., and Hearn, B.C., 1983, Paleomagnetism of the Late Cretaceous–early Tertiary north-central Montana alkalic province: Journal of Geophysical Research, v. 88, p. 10593–10609.

Dokka, R.K., 1983, Displacements on late Cenozoic strike-slip faults of the central Mojave Desert, California: Geology, v. 11, p. 305–308.

——1986, Patterns and modes of early Miocene crustal extension, central Mojave Desert, California, *in* Mayer, L., ed., Extensional tectonics of the southwestern United States—A perspective on processes and kinematics: Geological Society of America Special Paper 208, p. 75–95.

——1988, Synthesis of middle and late Cenozoic tectonic history of the Mojave Desert Block: Geological Society of America Abstracts with Programs, v. 20, p. 156.

Dokka, R.K., and Woodburne, M.O., 1986, Mid-Tertiary extensional tectonics and sedimentation, central Mojave Desert, California: Louisiana State University Publications in Geology and Geophysics, Tectonics and Sedimentation, v. 1, 55 p.

Fisher, R.A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, ser. A, v. 217, p. 295–305.

Garfunkel, Z., 1974, Model for the late Cenozoic tectonic history of the Mojave Desert, California, and its relation to adjacent regions: Geological Society of America Bulletin, v. 5, p. 141–188.

Glazner, A.F., 1986, Stratigraphy, structure, and potassic alteration of Miocene volcanic rocks in the Sleeping Beauty area, central Mojave Desert, California, *in* Nielson, J.E., and Glazner, A.F., eds., Cenozoic stratigraphy, structure and mineralization in the Mojave Desert (Geological Society of America Cordilleran Section meeting guidebook, field trip 5): Los Angeles, California State University, p. 51–63.

Glazner, A.F., Neilson, J.E., Howard, K.A., and Miller, D.M., 1986, Correlation of the Peach Springs Tuff, a large-volume Miocene ignimbrite sheet in California and Nevada, *in* Neilson, J.E., and Glazner, A.F., eds., Cenozoic stratigraphy, structure and mineralization in the Mojave Desert (Geological Society of America Cordilleran Section meeting guidebook, field trip 5): Los Angeles, California State University, p. 65–68.

Golombek, M., and Brown, L., 1986, Clockwise rotation of the northwest Mojave Desert [abs.]: EOS (American Geophysical Union Transactions), v. 67, p. 922.

——1988, Clockwise rotation of the western Mojave Desert: Geology, v. 16, p. 126–130.

Harlan, S.S., and Geissman, J.W., 1987, Preliminary paleomagnetic data from the mid-Cretaceous (?) pluton of Indian Springs, Cima volcanic field, eastern Mojave Desert, California [abs.]: EOS (American Geophysical Union Transactions), v. 68, p. 1253.

Hillhouse, J.W., and Wells, R.E., 1986, Assessment of rotations in the eastern Mojave Desert, California and Arizona, from paleomagnetism of the Peach Springs Tuff [abs.]: EOS (American Geophysical Union Transactions), v. 67, p. 922.

Jennings, C.W., Burnett, J.L., and Troxel, B.W., editors, 1963, Geologic map of California, Trona sheet: California Division of Mines and Geology, scale 1:250,000.

Luyendyk, B.P., Kamerling, M.J., and Terres, R.R., 1980, Geometric model for Neogene crustal rotations in southern California: Geological Society of America Bulletin, v. 91, p. 211–217.

Luyendyk, B.P., Carter, J.N., and Haston, R., 1985, Neogene clockwise rotations in southwestern California—The eastern Transverse Ranges, and the central Mojave Desert? [abs.]: EOS (American Geophysical Union Transactions), v. 66, p. 865.

MacFadden, B.J., Opdyke, N.D., and Woodburne, M.O., 1987, Paleomagnetism of the middle Miocene Barstow Formation, Mojave Desert, southern California—Magnetic polarity stratigraphy and tectonic rotation [abs.]: EOS (American Geophysical Union Transactions), v. 68, p. 291.

MacFadden, B.J., Woodburne, M.O., and Opdyke, N.D., 1988, Paleomagnetism and negligible tectonic rotation of the Miocene Hector and Barstow Formations, Mojave Desert, California: Geological Society of America Abstracts with Programs, v. 20, p. 176.

——1989, Paleomagnetism and Neogene clockwise rotation of the northern Cady Mountains, Mojave Desert of southern California: Journal of Geophysical Research (in press).

Miller, S.T., 1980, Geology and mammalian biostratigraphy of a part of the northern Cady Mountains, Mojave Desert, California: U.S. Geological Survey Open-File Report 80-878, 122 p.

Nason, G.W., Davis, T.E., and Stull, R.S., 1979, Cenozoic volcanism in the Newberry Mountains, San Bernardino County, California, *in* Armentrout, J.M., Cole, M.R., and Terbest, H., Jr., eds., Cenozoic paleogeography of the western United States: Society of Economic Paleontologists and Mineralogists, Pacific Coast Section, Pacific Coast Paleogeography Symposium 3, p. 89–95.

Nielson, J.E., Glazner, A.F., and Lux, D.R., 1988, Problems of dating the Peach Springs Tuff: Geological Society of America Abstracts with Programs, v. 20, p. 218.

Rogers, T.H., editor, 1969, Geologic map of California, San Bernardino sheet: California Division of Mines and Geology, scale 1:250,000.

Ross, T.M., Luyendyk, B.P., and Haston, R.B., 1987, Neogene tectonic rotations in the Cady Mountains, Mojave Desert, California: Geological Society of America Abstracts with Programs, v. 19, p. 824.

——1988, Paleomagnetic evidence for Neogene tectonic rotations in the central Mojave Desert, California: Geological Society of America Abstracts with Programs, v. 20, p. 226.

Valentine, M.J., Brown, L.L., and Golombek, M.P., 1987, Tectonic rotation of the Barstow area, central Mojave Desert, California, inferred from paleomagnetic data [abs.]: EOS (American Geophysical Union Transactions), v. 68, p. 1254.

——1988, Counterclockwise Cenozoic rotations of the Mojave Desert: Geological Society of America Abstracts with Programs, v. 20, p. 239.

Wells, R.E., and Hillhouse, J.W., 1989, Paleomagnetism and tectonic rotation of the lower Miocene Peach Springs Tuff—Colorado Plateau, Arizona to Barstow, California: Geological Society of America Bulletin (in press).

Wernicke, B., and Burchfiel, B.C., 1982, Modes of extensional tectonics: Journal of Structural Geology, v. 4, p. 105–115.

Zijderveld, J.D.A., 1967, A.C. demagnetization of rocks—Analysis of results, *in* Collinson, D.W., Creer, K.M., and Runcorn, S.K., eds., Methods of paleomagnetism: New York, Elsevier, p. 245–286.

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